1. Lab 09: IPv4, DHCP, and NAT

Lab Name: Lab 09: IPv4, DHCP, and NAT **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

Dynamic Host Configuration Protocol (DHCP) and Network Address Translation (NAT) are essential technologies for managing IPv4 addressing in modern networks. DHCP automates the assignment of IP addresses and network configuration parameters to devices, while NAT allows multiple devices on a private network to share a single public IP address, conserving the limited pool of globally routable IPv4 addresses. In this lab, you will configure both DHCP and NAT within your CML topology, observing how devices obtain IP addresses dynamically and how internal private addresses are translated for external communication.

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3. Objective / Purpose

To configure a router as a DHCP server for local clients and implement Port Address Translation (NAT Overload) to allow private IPs to reach an external network.

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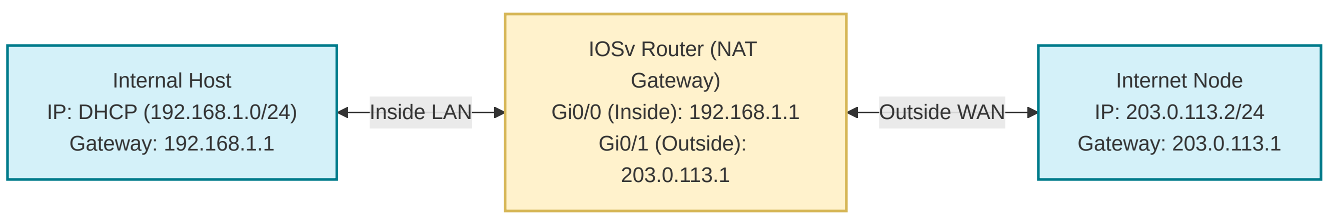
4. Topology Diagram



Details:

- 1x Router (NAT Gateway), 1x Internal Switch/Node, 1x External Node (representing the Internet).

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 1x IOSv Router, 2x Linux Nodes

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6. Configuration / Methodology

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	NAT Role
R1	G0/0 (Inside)	192.168.1.1	255.255.255.0	ip nat inside

R1	G0/1 (Outside)	203.0.113.1	255.255.255.0	ip nat outside

Key Configuration Snippets

```
! DHCP Config
ip dhcp pool LAN
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
! NAT Config
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

Narrative

We simulate an enterprise edge. The router dynamically assigns IPs to internal clients via DHCP. When clients attempt to ping the 'Internet' node, NAT overload translates their private source IP to the router's public outside interface IP.

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7. Verification & Testing

1. Output of `show ip dhcp binding` on R1.
2. Output of `show ip nat translations` after pinging the external node from the internal client.

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Successfully implemented DHCP for dynamic addressing and NAT overload to map a block of private IPv4 addresses to a single public IP address.

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10. What to Hand In

- The Lab YAML file with your name, date, name of the lab at the top of the file in comments.
- A PDF with annotated screen shots showing the lab running, as well as screen shots of Wireshark captures of the lab showing DHCP negotiation and NAT translation traffic.